

1. (16 points) Prove that every function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ can be computed by a circuit of size less than $1000 \cdot 2^n/n$.
2. (16 points) Prove that there exists an oracle A such that $P^A \neq NP^A$ but $NP^A \subseteq P^A/\text{poly}$.
3. (16 points) Show that for every $k > 0$ that PH contains languages whose circuit complexity is $\Omega(n^k)$.
4. (16 points) a) Let $\text{add}_n : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{n+1}$ be the function that takes two n -bit numbers and outputs their sum as an $(n + 1)$ -bit number. Show that add_n can be computed by a circuit of size $O(n)$.
(16 points) b) Show that $\text{Maj}_n : \{0, 1\}^n \rightarrow \{0, 1\}$, the majority function that outputs 1 if and only if at least half of the input bits are 1, can be computed by a circuit of size $O(n)$.
5. (21 points) Show that $\text{uniform NC}^1 \subseteq \text{L}$. Then show that $\text{PSPACE} \neq \text{uniform NC}^1$.